PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Tai Chi and other mind-body interventions for cancer-related fatigue: an updated systematic review and network meta-analyses protocol
AUTHORS	Fu, Hongjuan; Zhou, Hao; Tang, Yong; Li, Jie; Zhang, Da; Ding, Song-Yi; Huang, Qin-Wan; Wang, Chao

VERSION 1 – REVIEW

REVIEWER	Peter Herbison University of Otago, Preventive and Social Medicine
REVIEW RETURNED	17-Jun-2021

GENERAL COMMENTS	I think this is a good protocol. I think it could be improved by some minor changes. While this study is registered, there is a specific register for systematic reviews: PROSPERO at the University if York. I think this should be registered there as well. Are all Tai Chi programmes the same? It appears they are all going to be treated equally, but do they differ substantially in content, intensity, or frequency? In the Strengths and Limitations section to says that only the last 4 years is to be looked at. This is not mentioned in the methods. If this limitation is used then it should be justified. The authors say they will remove quasi-randomised trials. Unfortunately this is not possible. Many studies simply say they are randomised with no detail. Some of these will be quasi-randomised. So any attempt to remove quasi-randomised studies will only remove the better reported ones. The wording under the heading "Unit of Analysis" is unclear. The first bit of missing data talks about all sorts of missing data but the suggested remedy only applies for missing standard deviations. This should be cleared up. NMA is not expanded on first use in the text, but is a few sentences later. The authors use WMD (Weighted Mean Difference). This usage has been replaced now by simply MD. The abstract used the abbreviation SUCAR instead of SUCRA.

REVIEWER	Parco Siu University of Hong Kong

REVIEW RETURNED	21-Jun-2021

GENERAL COMMENTS	This Network Meta-analysis (NMA) protocol set to investigate the effects of tai chi and other mind-body interventions on cancer-related fatigue in patients who are undergoing or have completed cancer treatment. While the topic is interesting and follows the current research trends on the effectiveness of mind-body interventions, there seems to be already a solid body of evidence investigating this matter in a similar way as the reviewed study. As a result, the research novelty of this NMA is limited to the inclusion of more recent studies, but does not provide substantial improvements in the research methodology when compared to similar previously-published reviews. Furthermore, while the NMA is aimed to focus on mind-body interventions only (e.g. Tai-chi, yoga, qigong), the methods section clearly include interventions that do not fall within this category (i.e. resistance exercise, cognitive behavioral therapy, etc). With this regard, it is recommended to either change the aims of the study or adjust the inclusion criteria for the study search. In general, the language used throughout the manuscript is at times overly simple and contains a few grammatical and syntax errors. The below critiques are offered for author's consideration in the attempt to improve the overall study quality. Abstract The content provided in the abstract reflects the main points addressed in the main manuscript, although the language use is occasionally confusing and poorly structured. No major issues with this section were detected. ln 11-12: not clear what "which" refers to in this sentence. By the way the sentence is structured, it seems that it refers to cancer, in which case it is an underestimation of the effects that cancer has on our lives. If it refers to fatigue, I suggest rephrasing that sentence so that it is clearer what the subject of the sentence is. Page 5, Ln 12-13: not sure what "as the nonpharmacological therapy" means in this context. I suggest removing it to improve smoothness of the sentence. Ln 53: or standard care Ln 54: changes in fatigue Ln 54: not sure what effective means in this context. I suggest removing it and only keep "validated" Ln 55: to synthesize Introduction This section follows a logical order and touches the main points necessary to build a strong rationale on the topic. However, the reasons provided for conducting the study are rather weak. In fact, the authors mention that previous reviews are inconsistent to each other, while in reality they all support the use of tai-chi to alleviate cancer-related fatigue. It might be useful to bring a stronger rationale to conduct the present study. Page 8, Ln 20-21: not sure what "which" refers to in these circumstances. If it refers to CRF I suggest using another conjunction, such as "and" Ln 60: "some nonpharmacological interventions" sounds better and simpler Page 9, ln 33-34: again "which" is not the best option in this case. Ln 40: improvements Ln 50: meta-analyses Ln 60: have been

	Page 10, ln 4: the effectiveness Ln 4-8: I suggest rephrasing this sentence “to present an objective and comprehensive evidence based on the up-to-date resources” as it is overly complicated and not entirely correct Ln 14-21: following PICOS criteria, the objective fails to mention the main outcome of the study. It is not clear what “relative effects” means in this circumstance. I suggest rephrasing this and including a clear outcome. Methods This section is inclusive of all or most components of the methodology required for NMAs. However, there are a few points that need extra consideration. First, the inclusion criteria mention some types of interventions that cannot be defined as mind-body (i.e. cognitive therapy, graded exercise, strengthening exercise, physical training, resistance training, cognitive behavioral therapy). Therefore, the aims of the study need to be adjusted to include these interventions, or these must be excluded from the study. Second, the authors use different verbal tenses to describe the methodology of the study. Besides some specific circumstances, future tenses need to be used consistently across this section. Third, it is not entirely clear whether mind-body interventions will be compared to each other or to control comparisons only. In the “Types of studies” section, it is mentioned that you will include “RCTs that compared Tai Chi, other mind-body interventions against no intervention, placebo or standard care”. However, in the section named “Types of comparators” it is mentioned that “information will be obtained either from head-to-head or placebo RCTs” and that “participants in the control group should be treated with no intervention or all the interventions listed above”. If mind-body interventions will be compared to each other this has to be more clear from the text. Page 11, ln 18: I suggest rephrasing “during or post-treatment” to make it clearer that you are referring to whether the intervention was performed during or after cancer treatment Ln 40-47: the final sentence of this chapter needs to be rephrased as it is overly complex and not very clear. Page 12, ln 37: how will quality of life be assessed? Any specific questionnaires? Page 13, ln 11-14: manually searching Table 1: not sure why in #7 the parameters #3 and #6 are combined with “OR” instead of “AND” Page 14, ln 28: I do not understand this sentence “the full text of a 11 remaining articles will be obtained…”. Besides the grammatical incorrectness of the sentence, how do you already know how many studies will be included? Page 16, ln 50: to present Page 19, ln 43: according to Page 20, ln 14: we will rate

VERSION 1 – AUTHOR RESPONSE

We appreciate your comments: I think this is a good protocol. I think it could be improved by some minor changes.

Thanks for your positive comments.

Comment 1: While this study is registered, there is a specific register for systematic reviews: PROSPERO at the University if York. I think this should be registered there as well.

Response: We appreciate the reviewer's suggestion. This review protocol has been registered in PROSPERO with a registration number CRD42021244999. Please see the abstract and methods section.

Comment 2: Are all Tai Chi programmes the same? It appears they are all going to be treated equally, but do they differ substantially in content, intensity, or frequency?

Response: The reviewer has made a very good point. Tai Chi can be subdivided into Yang-style TCC, Chen-style TCC, Wu-style TCC, Sun-style TCC, 24 simplified TCC, or movements of TCC. Actually, the style, time, frequency, and duration of treatment of Tai Chi were variable. Currently, reviewer researchers treat different types of Tai Chi equally. ([1] Luo XC, et al. Effect of Tai Chi Chuan in Breast Cancer Patients: A Systematic Review and Meta-Analysis. Front Oncol. 2020 Apr 23;10:607. [2] Qin J, Chen Y, Guo S, You Y, Xu Y, Wu J, Liu Z, Huang J, Chen L, Tao J. Effect of Tai Chi on Quality of Life, Body Mass Index, and Waist-Hip Ratio in Patients With Type 2 Diabetes Mellitus: A Systematic Review and Meta-Analysis. Front Endocrinol (Lausanne). 2021 Jan 19;11:543627.). The reason may be that the original RCT researchers did not specify the type of Tai Chi reporting the research interventions. This brings difficulties to the review analysis of the research. In this study, all different types of Tai Chi will first be reviewed as the same intervention and compared with other control measures. Then, different types of Tai Chi will be analyzed again as different interventions. When our study is completed, we will suggest researchers to specify different types of Tai Chi in future reports, so as to observe whether they have different clinical effects.

Comment 3: In the Strengths and Limitations section to says that only the last 4 years is to be looked at. This is not mentioned in the methods. If this limitation is used then it should be justified.

Response: Thanks for your nice suggestion. We revised this sentence. Please see the strengths and limitations section.

Comment 4: The authors say they will remove quasi-randomised trials. Unfortunately, this is not possible. Many studies simply say they are randomised with no detail. Some of these will be quasi-randomised. So, any attempt to remove quasi-randomised studies will only remove the better reported ones.

Response: Very good point. We deleted this sentence.

Comment 5: The wording under the heading "Unit of Analysis" is unclear.

Response: Sorry for our confused introduction. We revised this sentence.

Comment 6: The first bit of missing data talks about all sorts of missing data but the suggested remedy only applies for missing standard deviations. This should be cleared up.

Response: Very good point. We provided a supplementary explanation for this part as you suggested.

Comment 7: NMA is not expanded on first use in the text, but is a few sentences later.

Response: Thanks for your nice suggestion. We corrected it as you suggested.

Comment 8: The authors use WMD (Weighted Mean Difference). This usage has been replaced now by simply MD.

Response: We appreciate the reviewer's constructive suggestion. According to the Cochrane Handbook for Systematic Reviews of Interventions, we will use standardized mean differences (SMD) instead of WMD. The standardized mean difference is usually used when included studies all report the outcome using different scales.
(https://training.cochrane.org/handbook/archive/v6.1/chapter-10#section-10-5).

Comment 9: The abstract used the abbreviation SUCAR instead of SUCRA.

Response: Sorry for the confusion. We corrected it as you suggested.

Response to REVIEWER 2:
We appreciate your comments: This Network Meta-analysis (NMA) protocol set to investigate the effects of tai chi and other mind-body interventions on cancer-related fatigue in patients who are undergoing or have completed cancer treatment. While the topic is interesting and follows the current research trends on the effectiveness of mind-body interventions, there seems to be already a solid body of evidence investigating this matter in a similar way as the reviewed study. As a result, the research novelty of this NMA is limited to the inclusion of more recent studies, but does not provide substantial improvements in the research methodology when compared to similar previously-published reviews.
Furthermore, while the NMA is aimed to focus on mind-body interventions only (e.g. Tai-chi, yoga, qigong), the methods section clearly include interventions that do not fall within this category (i.e. resistance exercise, cognitive behavioral therapy, etc). With this regard, it is recommended to either change the aims of the study or adjust the inclusion criteria for the study search.

Thanks for your insightful comments. We revised it in Methods section.

In general, the language used throughout the manuscript is at times overly simple and contains a few grammatical and syntax errors. The below critiques are offered for author's consideration in the attempt to improve the overall study quality.

Thank you very much for your great attention and patience to correct the grammar and misspelling mistakes. We followed your suggestion and got additional help from professional biomedical editing company to carefully proofread the manuscript and made corrections.

Abstract
The content provided in the abstract reflects the main points addressed in the main manuscript, although the language use is occasionally confusing and poorly structured. No major issues with this section were detected.
Comment 1: ln 11-12: not clear what "which" refers to in this sentence. By the way the sentence is structured, it seems that it refers to cancer, in which case it is an underestimation of the effects that cancer has on our lives. If it refers to fatigue, I suggest rephrasing that sentence so that it is clearer what the subject of the sentence is.

Response: Good suggestion. We revised it to avoid confusion.

Comment 2: Page 5, Ln 12-13: not sure what "as the nonpharmacological therapy" means in this context. I suggest removing it to improve smoothness of the sentence.

Response: Sorry for the confusion. We deleted it and proofread the manuscript.

Comment 3: Ln 53: or standard care

Response: Good point. We revised it.

Comment 4: Ln 54: changes in fatigue

Response: Sorry for the confusion. We corrected it.

Comment 5: Ln 54: not sure what effective means in this context. I suggest removing it and only keep "validated"

Response: Sorry for the mistake, we removed it.

Comment 6: Ln 55: to synthesize

Response: We revised it.

Introduction
This section follows a logical order and touches the main points necessary to build a strong rationale on the topic. However, the reasons provided for conducting the study are rather weak. In fact, the authors mention that previous reviews are inconsistent to each other, while in reality they all support the use of tai-chi to alleviate cancer-related fatigue. It might be useful to bring a stronger rationale to conduct the present study.
Comment 7: Page 8, Ln 20-21: not sure what "which" refers to in these circumstances. If it refers to CRF I suggest using another conjunction, such as "and"

Response: Sorry for the confusion. We rephrased this sentence as you suggested.

Comment 8: Ln 60: "some nonpharmacological interventions" sounds better and simpler

Response: Very good point. We replaced this phrase.

Comment 9: Page 9, ln 33-34: again "which" is not the best option in this case.

Response: We revises this sentence.

Comment 10: Ln 40: improvements

Response: We corrected it.

Comment 11: Ln 50: meta-analyses

Response: Thanks for your attention. We checked the Cochrane Handbook for Systematic Reviews of Interventions once again (https://training.cochrane.org/handbook/archive/v6.1). We modified it according to your suggestions.

Comment 12: Ln 60: have been

Response: Sorry for the mistake, we corrected it.

Comment 13: Page 10, ln 4: the effectiveness

Response: Good point. We revised it as you suggested.

Comment 14: Ln 4-8: I suggest rephrasing this sentence “to present an objective and comprehensive evidence based on the up-to-date resources” as it is overly complicated and not entirely correct

Response: Thanks for your suggestion. We have revised it to avoid confusion.

Comment 15: Ln 14-21: following PICOS criteria, the objective fails to mention the main outcome of the study. It is not clear what “relative effects” means in this circumstance. I suggest rephrasing this and including a clear outcome.

Response: Thanks for your insightful comments. We revised this sentence.

Methods
This section is inclusive of all or most components of the methodology required for NMAs. However, there are a few points that need extra consideration.
Comment 16: First, the inclusion criteria mention some types of interventions that cannot be defined as mind-body (i.e. cognitive therapy, graded exercise, strengthening exercise, physical training, resistance training, cognitive behavioral therapy). Therefore, the aims of the study need to be adjusted to include these interventions, or these must be excluded from the study.

Response: That is a very important issue raised by the reviewer. The National Center for Complementary and Alternative Medicine (NCCAM) is the Federal Government’s lead agency for scientific research on complementary and alternative medicine (CAM). NCCAM defines CAM mind and body interventions (MBIs) as follows: Acupuncture, Breath practices, Meditation, Guided imagery, Progressive, relaxation, Tai chi, Yoga, Spinal manipulation Massage therapy, Feldenkrais, method, Alexander technique, Pilates, Hypnosis, Trager psychophysical integration, Reiki, Healing touch, Qi gong, Craniosacral therapy, and Reflexology. We focus here on four types of MBIs that have received considerable research attention and are widely available to clinical and community populations: Tai Chi, Qigong (including Baduanjin, Yijinjing, and Wuqinxi), yoga, and meditation. (Bower JE, Irwin MR. Mind-body therapies and control of inflammatory biology: A descriptive review. Brain Behav Immun. 2016 Jan;51:1-11.) To constrain the considerable heterogeneity of MBTs, we will not include studies of acupuncture, massage, and spinal manipulation therapies in our review. Moreover, to determine whether the effects of Tai Chi and Qi gong are primarily due to physical activity and whether the effects of yoga are primarily due to stretching, exercise and stretching will be included as control group. Finally, we followed your suggestion to revise the interventions and comparators criteria.

Comment 17: Second, the authors use different verbal tenses to describe the methodology of the study. Besides some specific circumstances, future tenses need to be used consistently across this section.

Response: Thanks for your nice suggestion. We corrected it as you suggested.

Comment 18: Third, it is not entirely clear whether mind-body interventions will be compared to each other or to control comparisons only. In the “Types of studies” section, it is mentioned that you will include “RCTs that compared Tai Chi, other mind-body interventions against no intervention, placebo or standard care”. However, in the section named “Types of comparators” it is mentioned that “information will be obtained either from head-to-head or placebo RCTs” and that “participants in the

control group should be treated with no intervention or all the interventions listed above". If mind-body interventions will be compared to each other this has to be more clear from the text.

Response: We appreciate your constructive suggestions. In this research, mind-body interventions will be compared both with each other and with control comparisons. We revised these sentences to avoid confusion.

Comment 19: Page 11, ln 18: I suggest rephrasing "during or post-treatment" to make it clearer that you are referring to whether the intervention was performed during or after cancer treatment

Response: Thanks a lot for your attention. We rephrased this sentence.

Comment 20: Ln 40-47: the final sentence of this chapter needs to be rephrased as it is overly complex and not very clear.

Response: Sorry for our confused introduction. We deleted this sentence.

Comment 21: Page 12, ln 37: how will quality of life be assessed? Any specific questionnaires?

Response: We appreciate your constructive comments. Questionnaires will be used for assessment of QoL, such as the Functional Assessment of Cancer Therapy (FACT), European Organization for Research and Treatment of Cancer Quality of Life Questionnaire Core 30 (EORTC QLQ-C30), Quality of Life Questionnaire Breast Cancer Module 23 (EORTC QLQ-BR23), 36-Item Short Form Health Survey (SF-36), Quality of Life in Adult Cancer Survivors (QLCAS), World Health Organization Quality of Life Questionnaire, brief version (WHO QOL-BREF). We made supplementary explanation in the paper.

Comment 22: Page 13, ln 11-14: manually searching

Response: Sorry for the confusion. We revised this sentence.

Comment 23: Table 1: not sure why in #7 the parameters #3 and #6 are combined with "OR" instead of "AND"

Response: Sorry for the mistake. We revised the search strategy. Please see the supplement file.

Comment 24: Page 14, ln 28: I do not understand this sentence "the full text of a 11 remaining articles will be obtained…". Besides the grammatical incorrectness of the sentence, how do you already know how many studies will be included?

Response: Sorry for our confused introduction. This sentence should be "After this selection, the full text of all remaining articles will be obtained to determine which to include". Misleading caused by our Word format error. We revised the Word format of the article.

Comment 25: Page 16, ln 50: to present

Response: Very good point. We revised this sentence.

Comment 26: Page 19, ln 43: according to

Response: Thank you very much for your attention. We revised it.

Comment 27: Page 20, ln 14: we will rate

Response: We appreciate your constructive comments. We rephrased this sentence.

VERSION 2 – REVIEW

REVIEWER	Peter Herbison University of Otago, Preventive and Social Medicine
REVIEW RETURNED	12-Nov-2021

GENERAL COMMENTS	The authors have answered all my questions appropriately.

REVIEWER	Parco Siu University of Hong Kong
REVIEW RETURNED	03-Nov-2021

GENERAL COMMENTS	The authors have satisfactorily addressed my previous comments.